

# **VEDIC MATHEMATICS**

*Dissertation submitted to the University of Calicut in the partial fulfilment of  
the requirements for the award of the Degree of*

## **BACHELOR OF SCIENCE IN MATHEMATICS**



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### **CERTIFICATE**

This is to certify that the project report submitted by MOHAMED SHAFEEF T P (Reg No. SKAWSMT011), ARDRA JANARDHANAN (Reg No. SKAWSMT019) and SAMRUTHA T L (Reg No. SKAWSMT024) is a bonafide work carried out by them for the partial fulfilment of B.Sc. Mathematics course of University of Calicut during the year 2022-2025.

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## **DECLARATION**

We **MOHAMED SHAFEEF T P, ARDRA JANARDHANAN** and **SAMRUTHA T L** do hereby declare that the project entitled “**VEDIC MATHEMATICS**” submitted to the Department of Mathematics, Sree Keralavarma College Thrissur, is an original dissertation done by us under the supervision of **Dr.KRISHNA M**, Assistant Professor, Department of Mathematics, Sree Keralavarma College, Thrissur.

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# INTRODUCTION

Father of Vedic mathematics , Shri Bharati Krishna Tirtha Maharaj was born on 14th March 1884 in Puri, in Orissa. His original name was Venkatraman Shashri and his father's name was P.Narashima Shashri . In 1957, he wrote the presently available Vedic Maths book inspite of suffering from old age and ill health with weak eyesight. He ensured to continue with his studies and writing, deliver talks with enthusiasm and zeal. In 1958, he went on a tour to US and addressed audiences of Colleges,

Churches and other Institutions on World Peace , which was organized by Self Realization Fellowship of Los Angeles. In 1958, the typescript of Vedic Maths was sent for publication in US. Bharati Krishna Tirtha attained Mahasamadhi on 2nd February 1960. 5 years later, his book Vedic Maths was published

- Vedic mathematics is an ancient technique consisting of 16 sutras and sub sutras. It simplifies not only the Fundamental Arithmetic Operations, such as Addition , Subtraction , Multiplication and Division . Its application range from basic arithmetic to advanced fields like Digital Signal Processing , Cryptography , Machine learning and so on.

Vedic Maths help us to Improved mental calculation skills, Enhanced mathematical understanding and intuition, Increased speed and accuracy in mathematical calculations and Reduced reliance on calculators and other digital tools

Chapter 1 deals with the preliminary concepts used in the following chapters. Coming to Chapter 2, it explains the addition of numbers by using the vedic sutra puranapuranaabhyam. In Chapter 3, the subtraction of numbers using the vedic sutra, Nikhilam Navatascramam Dastah are demonstrated. Chapter 4 includes multiplication by various vedic sutras ( ekanyuena purvena ,Urdhva tiryabhyam ,Antyayordhasakepi).Chapter 5 describes the division of numbers using the vedic sutra Nikhilam. Chapter 6 explains the method to find out squares of numbers by the vedic sutras Dwanda yoga and ekadhikena purvena . And in Chapter 7, it is done with square roots used by the vedic sutra vilokanam

We can see there are 16 sutras and subsutras in vedic mathematics ;

## **VEDIC SUTRAS**

- EKADHIKENA PURVENA :-By one more than the previous one
- NIKHILAM NAVATASCRAMAM DASATAH :-All from nine and last from ten
- URDHVA TIRYAGBHYAM :-Vertically and Crosswise
- PARAVARTYA YOJAYET :- Transpose and Apply
- SUNYAM SAMYASAMUCCAYE :-The Summation equal to zero
- ANURUPYE SUNYAMANYAT :-If one is in Ratio, other one is zero
- SANKALANA VYAVAKALANABHYAM :-By Addition and subtraction
- PURANAPURANABHYAM :-By completion And Non Completion
- CALANA KALANABHYAM :-Sequential Motion
- VYASTISAMASTIH :- Whole as one and One as whole
- SEDANYAKENA CARAMENA :-Reminder by Last digit
- SOPANTYADVAYAMANTYAM :-Ultimate And twice the penultimate
- YAVADUNAM :-The deficiency
- EKANYUNENA PURVENA :- By one less than The previous one
- EKANYUNENA PURVENA :- By one less than The previous one
- GUNAKASAMUCCAYAH :-Collectivity of Multipliers

## **Vedic sub sutras**

- ANURUPYENA :-Proportionality
- SISYATE SESASAMJNAH:- Knowing remainder from reminder
- ADYAMADYENANTYA-MATYENA:-First by first and last by last
- KEVALAIH SAPTAKAMGUNYAT:- Only Multiple of seven
- VESTANAM :-Osculation \*
- YAVADUNAM TAVADUNAM :-Whatever be the deficiency, lessen it further
- YAVADUNAM TAVADUNIKRTYA VARGANACA YOJAYET :-Whatever the extent of its Deficiency, lessen it further to that extent And set up the square of deficiency
- ANTYAYOTDASAKEPI :- When the sum of The last digits is ten  
ANTYAYOREVA :- Only the last term
- SAMUCCAYAGUNITAH :-Sum of the Coefficients in the product
- LOPANASTHAPANABHYAM :-By elimination And retention
- VILOKANAM :-The product of the sum of The coefficient
- GUNITA SAMUCCAYAH SAMUCCAYA GUNITA :-The product of the sum of the coefficients in the factor is equal to the Sum of the coefficients in the product
- DWANDWAYOGAH :-Duplex combination
- Dhvajank :- Flag digit

## **CONTENTS**

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# CHAPTER 1

## PRELIMINARIES

**Definition 1.1:-** Here addition is performed by the vedic sutra puranapuranabhyam. In puranapuranabyam we use the compliment method which means the sum of last digits equal to 10 or 100. For example  $37+83 = 30+80+10=120$

**Definition 1.2:-** Coming to the case of subtraction we use the Vedic sutra Nikhilam Navatascharam Dasatah which means All from nine and last from ten . While subtracting when the upper digit > lower digit ,normal subtraction is done . Otherwise we take the compliment of the differences.

Numbers : 0 1 2 3 4 5 6 7 8 9

Compliment. : 10 9 8 7 6 5 4 3 2 1

**Definition 1.3:-** Doing Multiplication ,if the sum of unit digit is 10 and remaining digits are same then we can use the Vedic sutra Antyayordhashakepi . In this case the multiplication is done by multiplying the unit digits we get the RHS and for LHS multiply by  $n(n+1)$  . [ remaining digit x remaining digit +1]

**Definition 1.4:-** While Multiplying with 9's we can use the Vedic sutra ekanyuena purvena (means, by one less than the previous one) . Here we discuss about 3 different cases

1. When the number of digits in the multiplicand and number of 9's in the multipliers is same.
2. When the number of 9's in the multipliers is more than the number of digits in the multiplicand.
3. When there is less number of 9's in the multipliers than the number of digits in the multiplicand.

[Multiplicand :- The multiplicand is the number being multiplied. It is the number that is being repeated or added to itself a certain number of times.]

[Multiplier :- The multiplier is the number by which the multiplicand is being multiplied. It is the number that tells us how many times to repeat or add the multiplicand.]

**Definition 1.5 :-** The Vedic sutra Urdhva Tiryagbhyam ( vertically and cross wise) is an universally accepted multiplication technique. While using this method we can avoid the several steps in normal multiplication. Here multiplication of two digit numbers consists 3 steps ( vertically -> cross wise -> vertically ) and add numbers we get the final result and multiplication of 5 digits consist 5 steps .

**Definition 1.6 :-** Multiplication by 11,111 follows the same pattern . It's all depends how many 1's are there. Multiplication of 11 done by adding nearest 2 digits from right to left ( put a zero in the RHS and LHS side). Multiplication of 111 done by adding nearest 3 digits from right to left (put 2 zeros on both RHS and LHS side ).

**Definition 1.7 :-** Multiplication by powers of 5.

(5,25,125,625....) which has a general form

$$5^n = 10^n \div 2^n$$

Which means multiplication by 5 = multiply the digit by 10 and divide by 2

Multiplication by 25= multiply by 100 and divide by 4

Multiplication by 125= multiply by 1000 and divide by 8.

**Definition 1.8 :-** Division we use Nikhilam method in this method set 3 columns . Column 1 divisor and it's compliment Column 2 for quotient and Column 3 for reminder . This method is like to synthetic division .

**Definition 1.9 :-** Finding squares of numbers using duplex method . This method is useful for every numbers .for this method find the duplex of the numbers and add it from right to left .

**Definition 1.10:-** For getting the numbers whose Squares ending with 5 , we can use the Vedic sutra ekadhikena purvena ( means by one more than the previous one) .the general form is given by  $(n5)^2 = n(n+1) \mid 25$ .

**Definition 1.11:-** For getting Square root of a number ,we can use vilokanam method . For this we need to know the unit digit of perfect squares which are 0 ,1 , 4 ,5 ,6, 9.

## CHAPTER 2

### ADDITION

Here addition is performed by the vedic sutra puranapuranaabhyam.

|            |    |   |   |   |   |   |   |   |   |   |
|------------|----|---|---|---|---|---|---|---|---|---|
| Number     | 0  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Complement | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |

#### **Rule :-**

- 1) Spot those numbers which if paired would result in a rounded result. The above complement table will help in making such pairs immediately.
- 2) Rearrange the numbers and add as per the pairing of numbers.

**Example 1:-** Add:  $26 + 59 + 394 + 66 + 11 + 14 = ?$

Step 1: Here, the complete observation shows us that  $26 + 14$  are likely to yield a rounded result. The same would be the case if  $59$  and  $11$  were paired. Moreover, the pair  $394$  and  $66$  yield a rounded result.

Step 2: Rearrange the pair and add as per the pairing done above.

$$= (26 + 14) + (59 + 11) + (394 + 66)$$

$$= 40 + 70 + 460$$

$$= (40 + 460) + 70$$

$$= 500 + 70$$

$$= 570$$

Hence,  $26 + 59 + 394 + 66 + 11 + 14 = 570$

## CHAPTER 3

### Subtraction

Here subtraction is performed by the Vedic Sutra Nikhilam Navatascramam Dastah ( All from 9 and last from 10 ).

**Rule :-**

- 1) When the upper digit is  $>$  the lower digit ,normal subtraction is done.
- 2) In case the upper digit  $<$  lower digit , we take the compliment of the differences as shown in the compliment table .

The compliment of the last digit is taken from 10 and the compliment of rest of the digits are taken from 9.

|            |    |   |   |   |   |   |   |   |   |   |
|------------|----|---|---|---|---|---|---|---|---|---|
| Numbers    | 0  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Compliment | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |

- 3) When you arrive at the stage where there is no need to take the compliment, subtract one extra from that column.

**Example 1:**  $854 - 569$

solution :  $854 -$

$$\begin{array}{r} 569 \\ \hline 285 \end{array}$$

Here are the explanation

Step 1: Since  $4 < 9$  , we take the compliment of the differences of the digits . This compliment is taken from 10 . The difference is 5 and its compliment is also 5 , so write 5 at unit place .

Step 2 : Again  $5 < 6$  , so we take the compliment of the differences of the digits , but this time the compliment is taken from 9 instead of 10 .As the sutra suggests( ALL FROM 9 AND LAST FROM 10) .The difference is  $6 - 5 = 1$  and its compliment from 9 is  $9 - 1 = 8$ . Write 8 at ten's place

. Step 3 : now  $8 > 5$  , so we don't need to take the compliment for the third column . As we are now out of the compliment , subtract 1 more in this column.  $8 - 5 - 1 = 2$  , write 2 at hundred's place .

## Chapter 4

# MULTIPLICATION

Here multiplication is performed by various vedic sutras and which are given below .

- 1) **Vedic sutra** : Ekanyuena Purvena ( by one less than the previous one ) This sutra work only under three conditions .

**Case 1)** When the number of digits in the multiplicand and number of

9's in the multipliers is same.

**Rule :-**

- a) Subtract 1 from the multiplicand and write the result in LHS.  
b) Subtract the multiplicand by nines ( Nikhilam Navatascramam Dastah) and write the result in RHS Example :  $6543 \times 9999 = ?$

Solution : in LHS  $6543 - 1 = 6542$  and RHS  $9999 - 6543 = 3457$  , so the answer is 34576542.

**Case 2)** when the number of 9's in the multipliers is more than the number of digits in the multiplicand.

**Rule :-**

- 1) The same procedure is applied as in case 1.

**Example** :  $456 \times 9999$  Ans) In LHS  $456 - 1 = 455$  and RHS  $9999 - 0455$

= 9544 so the answer is 4559544

Q)  $56892 \times 9999999$  Ans) In LHS  $56892 - 1 = 56891$  and RHS  $9999999 - 0056891 = 9943108$  , so the answer is 568919943108.

Here we can see that we add zeros at the beginning of the multiplicand to make the number of digits in the multiplicand and multiplier side become equal. So we can apply case 1 method.

**Case 3)** when there is less number of 9's in the multipliers than the number of digits in the multiplicand.

**Rule :-**

1) Add as many zeros the number of 9's to the multiplicand 2) Subtract the original multiplicand from the number obtained in step 1.

Example :  $1564 \times 99$

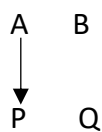
Ans) \* since there are two 9s ,put two zeros at the end of 1564, making it 156400.

\* subtract (1564 original multiplicand ) from 156400 =154836

**2)Vedic sutra : Urdhva Tiryagbhyam**

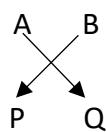
**a) Multiplication of two digits**

step 1



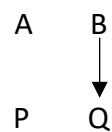
$A \times P$

step 2



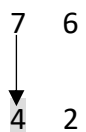
$(A \times Q) + (P \times B)$

step 3

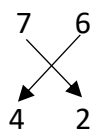


$B \times Q$

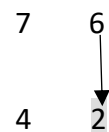
**Example 1:** Multiply 76 by 42



28



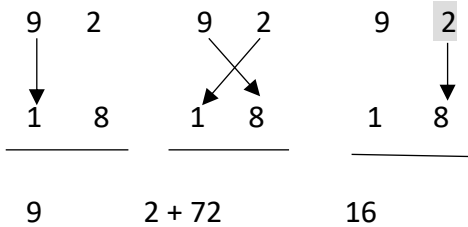
$24 + 14$



12

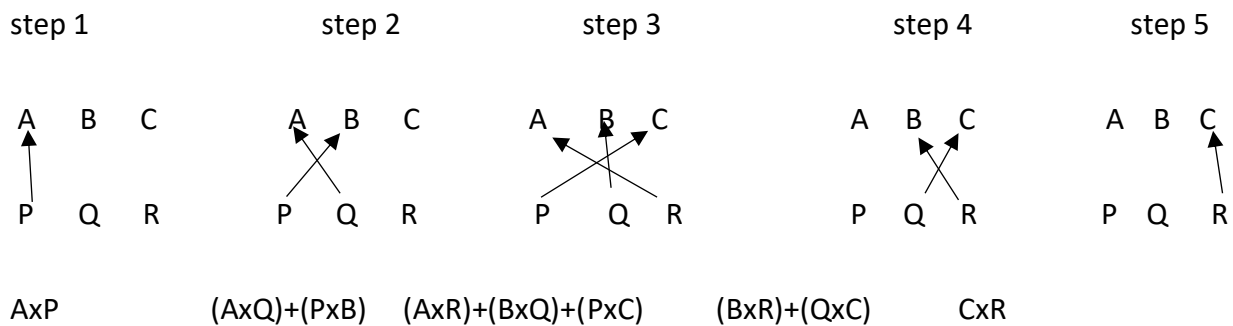
Arranging the number and adding them from right to left, taking only one digit at a time , we get the Final result ,  $28 \mid 38 \mid 12 = 3192$

**Example 2:**  $92 \times 18$

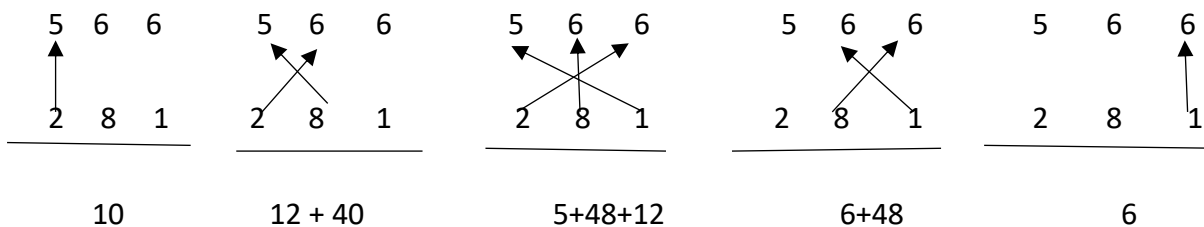


Final result =  $9 \mid 74 \mid 16 = 1656$

### ***b) Multiplication of three digits***



**Example:-1**



$10 \mid 52 \mid 65 \mid 54 \mid 6 = 159046$

## ***Multiplication by 11***

### ***Rule :-***

1. Place the number to be multiplied by 11 in a bracket and put zeros on either side.
2. Start adding the two numbers at a time from right to left. If the sum of two numbers in any case exceeds 10 , the digit at the tenth place shall be carried over to the next sum, as is unusually done in simple addition.

**Example1:-** Multiply 3251 by 11

1. Place the number in a bracket and put zeros on either side.

$$0 \ ( \ 3 \ 2 \ 5 \ 1 \ ) \ 0$$

2. Add the digit from the right to left.

$$0 + 3 \mid 3 + 2 \mid 2 + 5 \mid 5 + 1 \mid 1 + 0 = 3 \ 5 \ 7 \ 6 \ 1$$

Hence,  $3251 \times 11 = 35761$

**Example2:-** Multiply 4876254 by 11

1. Place the number in a bracket and put zeros on either side.

$$0 \ ( \ 4 \ 8 \ 7 \ 6 \ 2 \ 5 \ 4 \ ) \ 0$$

2. Add the digit from the right to left.

$$0 + 4 \mid 4 + 8 \mid 8 + 7 \mid 7 + 6 \mid 6 + 2 \mid 2 + 5 \mid 5 + 4 \mid 4 + 0$$

$$= 4 \mid 12 \mid 15 \mid 13 \mid 8 \mid 7 \mid 9 \mid 4$$

$$= 4 \mid 12 \mid 16 \mid 3 \mid 8 \mid 7 \mid 9 \mid 4$$

$$= 4 \mid 13 \mid 6 \mid 3 \mid 8 \mid 7 \mid 9 \mid 4$$

$$= 5 \ 3 \ 6 \ 3 \ 8 \ 7 \ 9 \ 4$$

Hence,  $4876254 \times 11 = 53638794$ .

### ***Multiplication by 111***

#### ***Rule:-***

Multiplication with 111 involves the same pattern as the multiplication with 11 with a slight difference.

1.Place the number to be multiplied inside a bracket and two zeros on either sides of bracket.

2.Add from right to left taking the sum of three digits at a time.

3.If the sum of the digits exceeds 10, the digit at the ten's place will be carried over to the next sum. Example:-Multiply 34 by 111 place the number in a bracket and put 2 zeros on either side.  $00(34)00$  keep adding three digits from the right at a time  $= 0+0+3 \mid 0+3+4 \mid 3+4+0 \mid 4+0+0 = 3774$   
(The same pattern follows on 1111,11111,etc.)

### ***Multiplication by 5***

#### ***Rule:-***

Put one zero to the right of the number and divide it by 2.

Example:-Multiply 5986 by 5 Place one zero after 5968, making it 59680 Divide it by 2

$59680/2=29930$  Hence 29930 is the product of 5968 and 5

## ***Multiplication by 125***

### ***Rule:-***

Put three zeros to the right of the multiplicand and divide it by 8.

**Example1:-**Multiply 624 by 125 Place three zeros after 624, making it 624000.

Divide it by 8

$624000/8=78000$  Hence 78000 is the product of 624 and 125.

**Example2:-**Multiply 24376 by 125 Place three zeros after 24376, making it

24376000 Divide it by 8

$24376000/8=3047000$  Hence 3047000 is the product of 24376 and 125

## ***Multiplication by 625***

### ***Rule:-***

Put four zeros (0000) to the right of the numbers and divide it by 16 Example:-Multiply 428 by 625 Place 4 zeros after 428, making it 4280000 Divide it by 16

$4280000/16=267500$  Hence, 267500 is the product of 428 and 625

## ***Multiplication by 51***

### ***Rule:-***

1:-Place the multiplicand on the right side. The right side should contain only two digits. If the number of digits at the right side is more than 2, the leftmost digit will be carried to the next column on the left side.

2:-On the left, place half of the multiplicand.

3:-In case the half of the multiplicand is fractional, add 50 to the digit placed on the right side.

This case will occur if the digit of the multiplicand is odd. Example:-Multiply 124 by 51

124

x 51

Place 124 at the right side.

124

X 51

---

/124

Place half of the multiplicand to the left, ie,  $124/2=62$ . The excess digit from the right side will get transferred to the left side.

124

X 51

---

/124 =63/24

Hence 6324 is the product of 124 and 51

***WHEN THE SUM OF UNIT'S PLACE IS 10 AND REST OF THE DIGITS ARE SAME  
(Antyayordhashakepi)***

***Rule:-***

1:-Multiply the unit digit those sum is 10 and place it on the right side.

2:-Increase the multiplicand by 1 and then multiply it with the original multiplier. Put this result to the left

**Example:-**multiply 46 by 44

Here the sum of the unit digit is 10 and the digit in the ten's place in the multiplier and multiplicand are also same.

4 6

X 4 4    ( Multiply the encircle digits and wright it to the right side. )

---

/24

Increase the multiplicand by 1 and multiply it with the original multiplier (4+1)=

56

X 4 6

20/24 Hence 2024 is the product of 46 and 44

### **Example 2**

71

X 79

56 /09

Hence the product of 71 x 79 = 5609

**WHEN THE SUM LAST TWO DIGITS IS 100 AND THE REST OF THE DIGITS AT HUNDRED'S PLACE ARE SAME .**

**Example:-**

Multiply 103 by 197

1:-Multiply the number (unit and ten's place digit ) of the multiplicand and multiplier whose sum is 100. The product should be of 4 digits. If the product in RHS is less than 4 digits, place zero at the extreme left.

1 0 3

X 1 9 7

/0291 (Since  $03 \times 97 = 291$  contains only 3 digits, one zero has been placed at the extreme left, making it 0291. As explained above, the right side of the product should contain 4 digits.)

2:-Increase the multiplicand (digit at hundred's place) by 1 and multiply it with the number placed at the hundred's place in the multiplier.

(1+1) =

2 20 3

X 1 9 7

2/0291

Hence 20291 is the product of 103 and 197

## CHAPTER 5

### DIVISION

Here division is performed by the Vedic sutra Nikhilam Navatasramam Dastah

***Rule:-***

1:-Take a base (in the power of ten) nearest to the divisor and write its complement below the original divisor, in the divisor column. Complement= Base – divisor. The complement can be carried out easily by the Nikhilam Navataha Charmam Dastah method as explained in the subtraction chapter. Write the complement of the divisor below it.

2:-Separate the extreme right digit of the dividend by drawing a slash equal to the number of digits in the divisor. This block is known as the remainder block and the left block is known as the quotient block.

3:-The number of digits to be placed in the remainder column should be equal to the number of zeros in the base.

4:-Carry down the first digit of divisor in the first column. This gives you the first digit of the quotient. Multiply the quotient digit by the complement, and place it in the dividend column; next to the first digit of the dividend.

5:-Write mechanically the sum of the digits of the second column to get the second digit of the quotient.

6:-Repeat the process until you get a number in the remainder column. If the number is greater than the divisor, continue the same process in the remainder block until the digit in the remainder column is less than that of the original divisor

### Case 1

**When the remainder is less than the divisor**

**Example :-10025 / 88**

1) Here divisor is 88 , base is 100 , complement =  $100 - 88 = 12$

|            | Column 1 | column 2(Q) | column 3(R) |
|------------|----------|-------------|-------------|
| Divisor    |          |             |             |
| Divisor    | 8 8      | 1 0 0       | 2 5         |
| Complement | 1 2      |             |             |

2) Arrange the digit in the columns as shown above, separating quotient and remainder. Since there are two zeros in the base, the remainder column will have the two right-most digit of the dividend.

3) Carry down 1 of column 2 (the first digit of the dividend). This is the first digit of the quotient.

4) Multiply the first digit of the quotient with the complement and place it in the dividend column, next to the first digit of the dividend.  $12 \times 1 = 12$  is placed below 0.

|     |        |           |            |
|-----|--------|-----------|------------|
|     | Column | column(Q) | column (R) |
|     | 8 8    | 1 0 0     | 2 5        |
| C = | 1 2    | 1 2       | - -        |
|     |        | 1 1       |            |

5) Carry down the sum of the circled digits; this will give you the second digit of the quotient. The second digit of the quotient is  $0 + 1 = 1$ .

6) Multiply the second digit of the quotient with the complement and place it in the dividend column, next to the second digit of the dividend.  $12 \times 1 = 12$  is placed below the second zero.

|     |          |             |             |
|-----|----------|-------------|-------------|
|     | Column 1 | Column 2(Q) | Column 3(R) |
|     | 8 8      | 1 0 0       | 2 5         |
| c = | 1 2      | 1 2         | - -         |
|     |          | 1           | 2           |
|     |          | 1 1 3       |             |

7) Carry down the sum of the second circled digits; this will give you the third digit of the quotient. The third digit of the quotient is  $0+2+1=3$

8) Multiply the complement 12 with the third quotient(3) and write it below the fourth digit of the reminder.  $12 \times 3 = 36$  is placed below 2 in the column 3.

|     |          |             |             |
|-----|----------|-------------|-------------|
|     | Column 1 | Column 2(Q) | Column 3(R) |
|     | 8 8      | 1 0 0       | 2 5         |
| c = | 1 2      | 1 2         | - -         |
|     |          | 1           | 2 -         |
|     |          |             | 3 6         |
|     |          | 1 1 3       | 8 1         |

9) Sum up the digits of column 3 to get the reminder. The above process repeats until the digit thus obtained in the reminder column is less than the original divisor. Here the sum of the digits of column 3 = 81 and  $81 < 88$  (divisor), hence the quotient=113 and reminder=81.

**When the reminder is more than the divisor**

**Example:-** Divide 10312 by 87

Base=100, Complement=100-87=13

|     |          |             |             |           |
|-----|----------|-------------|-------------|-----------|
|     | Column 1 | Column 2(Q) | Column 3(R) |           |
|     | 8 7      | 1 0 3       | 1 2         |           |
| c = | 1 3      | 1 3         | - -         | (13X1=13) |
|     |          | 1           | 3           | (13X1=13) |
|     |          |             | 9 1         | (13X7=91) |
|     |          | 1 1 7       | 13 3        |           |

Since the remainder 133<87, add the complement 13 in the remainder column.

|     |          |             |             |           |
|-----|----------|-------------|-------------|-----------|
|     | Column 2 | Column 2(Q) | Column 3(R) |           |
|     | 8 7      | 1 0 3       | 1 2         |           |
| C = | 1 3      | 1 3         | - -         | (33X1=13) |
|     |          | 1           | 3           | (13X1=13) |
|     |          |             | 9 1         | (13X7=91) |
|     |          | 1 1 7       | 1 33        |           |
|     |          |             | + 13        |           |

1 1 7 ← 1 46

1 1 8 46

Since the base is 100, the number in the remainder column should be less than 100; therefore the left-most 1 of the remainder column will be transferred to the quotient column. Thus quotient=117 and remainder=46.

## **CHAPTER 6**

### **SQUARE**

The Dwandwa-Yoga or Duplex method of squaring is one of the best squaring methods in vedic mathematics. By using this sutra, we can find the square of any number, of any length, with comfort ease in one line. We can find the square of any number mentally. This is unique in the sense that it has universal application. Let us denote the duplex of a number by D.

1. Duplex of 1 digit number = square of that number

$$D(a) = a^2$$

$$\text{Duplex of } 2 = 2^2 = 4 \quad \text{duplex of } 6 = 6^2 = 36$$

2. Duplex of 2 digit number = 2 X (Product of digits)

$$D(ab) = 2ab$$

$$\text{Duplex of } 24 = 2 \times (2 \times 4) = 16 \quad \text{Duplex of } 76 = 2 \times (7 \times 6) = 84$$

3. Duolex of 3 digit number = 2 X (1<sup>st</sup> digit X 3<sup>rd</sup> digit ) + ( square of middle digit )

$$D(abc) = 2ac + 2bc$$

$$\text{Duplex of } 126 = 2 \times (1 \times 6) + 2^2 = 16$$

$$\text{Duplex of } 478 = 2 \times (4 \times 8) + 7^2 = 113$$

4. Duplex of 4 digit num,ber = 2 X (1<sup>st</sup> digit X 4<sup>th</sup> digit ) + 2 X ( 2<sup>nd</sup> digit X 3<sup>rd</sup> digit )

$$D(abcd) = 2ad + 2bc$$

$$\text{Duplex of } 2468 = 2 \times (2 \times 8) + 2 \times (4 \times 6) = 80$$

$$\text{Duplex of } 4567 = 2 \times (4 \times 7) + 2 \times (5 \times 6) = 116$$

5 Duplex method of 5 digit number =  $2 \times (1^{\text{st}} \text{ digit} \times 5^{\text{th}} \text{ digit}) + 2 \times (2^{\text{nd}} \text{ digit} \times 4^{\text{th}} \text{ digit}) + (\text{middle digit})^2$

$$D(abcde) = 2ae + 2bd + c^2$$

$$\text{Duplex of } 16289 = 2 \times (1 \times 9) + 2 \times (6 \times 8) + 2^2 = 118$$

$$\text{Duplex of } 50406 = 2 \times (5 \times 6) + 2 \times (0 \times 0) + 4^2 = 76$$

6. Duplex of 6 digit number =  $2 \times (1^{\text{st}} \text{ digit} \times 6^{\text{th}} \text{ digit}) + 2 \times (2^{\text{nd}} \text{ digit} \times 5^{\text{th}} \text{ digit}) + 2 \times (3^{\text{rd}} \text{ digit} \times 4^{\text{th}} \text{ digit})$

$$D(abcdef) = 2af + 2be + 2cd$$

$$\text{Duplex of } 320416 = 2 \times (3 \times 6) + 2 \times (2 \times 1) + 2 \times (0 \times 4) = 40$$

$$\text{Duplex of } 125673 = 2 \times (1 \times 3) + 2 \times (2 \times 7) + 2 \times (5 \times 6) = 94$$

7. Duplex method of 7 digit number =  $2 \times (1^{\text{st}} \text{ digit} \times 7^{\text{th}} \text{ digit}) + 2 \times (2^{\text{nd}} \text{ digit} \times 6^{\text{th}} \text{ digit}) + 2 \times (3^{\text{rd}} \text{ digit} \times 5^{\text{th}} \text{ digit}) + (4^{\text{th}} \text{ digit})^2$

$$D(abcdefg) = 2ag + 2bf + 2ce + d^2$$

$$\text{Duplex of } 2356214 = 2 \times (2 \times 4) + 2 \times (3 \times 1) + 2 \times (5 \times 2) + 6^2 = 78$$

$$\text{Duplex of } 1035962 = 2 \times (1 \times 2) + 2 \times (0 \times 6) + 2 \times (3 \times 9) + 5^2 = 65$$

Now we need to write the number in groups. The following pattern will help to grouping the numbers.

$$(11)^2 = 121$$

$$(111)^2 = 12321$$

$$(1111)^2 = 1234321$$

$$(11111)^2 = 123454321$$

$$(111111)^2 = 12345654321$$

$$(1111111)^2 = 1234567654321$$

The grouping of  $(24)^2$  will follow the pattern of 1 2 1 of  $(11)^2$  the group for are

$\underbrace{\quad}_{D(2)}$

1 digit

$\underbrace{\quad}_{D(24)}$

2 digit

$\underbrace{\quad}_{D(4)}$

3 digit

the group of numbers for 2456 are -

$\underbrace{\quad}_{D(2)}$

1 digit

$\underbrace{\quad}_{D(24)}$

2 digit

$\underbrace{\quad}_{D(245)}$

3 digit

$\underbrace{\quad}_{D(2456)}$

4 digit

$\underbrace{\quad}_{D(456)}$

3 digit

$\underbrace{\quad}_{D(56)}$

2 digit

$\underbrace{\quad}_{D(6)}$

1 digit

**Example :** find the square of 8254

The group of number for 8254 are

8, 82, 825, 8254, 254, 54 and 4 .

Duplex of 8 =  $8^2 = 64$

Duplex of 82 =  $2 \times 8 \times 2 = 32$

Duplex of 825 =  $2 \times 8 \times 5 + 2^2 = 84$

Duplex of 8254 =  $2 \times 8 \times 4 + 2 \times 2 \times 5 = 84$

Duplex of 254 =  $2 \times 2 \times 4 + 3^2 = 41$

Duplex of 54 =  $2 \times 5 \times 4 = 40$

Duplex of 4 =  $4^2 = 16$

Arrange the value of duplex as follows –

$$64 \mid 2 \mid 4 \mid 4 \mid 1 \mid 0 \mid 6 \mid$$

$$= 68128516$$

$$\text{Hence } (8254)^2 = 68128516$$

### **Squares of numbers ending with 5**

For the number 25, the last digit is 5 and previous digit is 2. Hence, consider the example  $25^2$ .

Here the number is 25. We have to find out the square of the number.

‘One more than previous one’, that is,  $2+1=3$ .

The sutra, in this context, gives the procedure ‘to multiply the previous digit 2 by one more than itself, that is, by 3’.

It becomes L.H.S (left hand side) of the result, that is,  $2 \times 3 = 6$ .

The R.H.S (right hand side) of the result is  $5^2$ , that is, 25.

$$\text{Thus } 25^2 = 2 \times 3 / 25 = 625$$

In the same way,

$$1) \quad 35^2 = 3 \times (3+1) / 5^2$$

$$= 3 \times 4 / 25 = 1225$$

$$2) \quad 65^2 = 6 \times 7 / 25$$

$$= 4225$$

$$3) \quad 135^2 = 13 \times 14 / 25 = 18225$$

## CHAPTER 7

# SQUARE ROOT

Before we move to the vedic methodology to find the square root of a number, let us understand the following fundamental rules

1. A perfect square ends in 0,1,4,5,6 and 9
2. A number is not a perfect square if it ends with 2,3,7 or 8
3. If the given number has  $n$  digits then its square root will have  $n/2$  digits if  $n$  is even else  $(n+1)/2$  digit if  $n$  is odd

| N  | $N^2$ | Last digit of $N^2$ |
|----|-------|---------------------|
| 1  | 1     | 1                   |
| 2  | 4     | 4                   |
| 3  | 9     | 9                   |
| 4  | 16    | 6                   |
| 5  | 25    | 5                   |
| 6  | 36    | 6                   |
| 7  | 49    | 9                   |
| 8  | 64    | 4                   |
| 9  | 81    | 1                   |
| 10 | 100   | 0                   |

From the above table we conclude that :-

1. A complete square ending in 1 must have either 1 or 9 as the last digit of the square root.
2. A square ending in 4 must have 2 or 8 as the last digit of the square root.
3. A square ending in 6 must have 4 or 6 as the last digit of the square root.

**Example1:-** Find the square root of 2116.

1. Make a group of two from right.

21

16

2<sup>nd</sup> group

1<sup>st</sup> group

2. The unit digit of the 1<sup>st</sup> pair is 6 , so the square root ends in 4 or 6 (from the above table).
3. Now look at the second group. Since  $16 < 21 < 25$  , the digit at the ten's place = 4
4. Now we have two options  $\sqrt{2116} = 44$  or 46.

We know that  $45^2 = 2025$  ,  $2116 > 2025$  , therefore the desired square root will be more than 45.

Hence ,  $\sqrt{2116} = 46$

Example 2:- Find the square root of 5184

1. Make a group of two from the right

51

84

2<sup>nd</sup> group

1<sup>st</sup> group

2. The unit digit of 1<sup>st</sup> pair is 4, so the square root ends in 2 or 8.
3. Now look at the second group. Since  $49 < 51 < 84$ , the digit at the ten's place = 7.
4. Now we have two options  $\sqrt{5184} = 72$  or 78

We know that  $75^2 = 5625$  , since  $5184 < 5625$  , therefore the desired square root will be less than 75

Hence,  $\sqrt{5184} = 72$

## CONCLUSION

Vedic Maths is an ancient Indian mathematical system that offers a unique and efficient approach to mathematical problem-solving. Through this project, we have explored the principles and techniques of Vedic Maths, including the use of sutras, digit multiplication, and squaring. We can do problems fastly and efficiently while applying Vedic maths .We have also seen how Vedic Maths can be applied to various mathematical operations, such as addition, subtraction, multiplication, and division.

## REFERENCES

- VEDIC MATHEMATICS
  - JAGADGURU SWAMI SRI BHARATI KRSNA  
TIRTAJI MAHARAJA
- THE ESSENTIALS OF VEDIC MATHEMATICS
  - RAJESH KUMAR THAKUR